FIN2020 Homework 8

Problem 1

Think of a simple exchange economy with two agents that have the following utility functions:

$$U_1(\cdot) = 0.25c_0 + 0.5E[\ln(c_1(\theta))]$$

$$U_2(\cdot) = c_0 + E[\ln(c_1(\theta))]$$

where θ denotes the two equally likely states at date 1. The endowments are

	Date $t = 0$	*Date* $t = 1$	
		$\theta = 1$	$\theta = 2$
Agent 1	2	2	4
Agent 2	4	4	2

a. Describe the Pareto optimum or the set of Pareto optima. Is it unique? Is perfect risk sharing achieved? Why or why not? Answer the same question if the state 2 endowments were 5 for agent 1 and 3 for agent 2.

b. Agents can trade with two Arrow–Debreu securities denoted by Q [with payoffs (1,0)] and R [with payoffs (0,1)]. Calculate the competitive equilibrium allocation. Is it Pareto optimal? Discuss intuitively its characteristics (the determinants of the prices of the two securities and the post-trade allocation).

c. Suppose that both securities can be traded at the beginning and there are no short-selling constraints. A firm has zero cost of introducing one unit of either security, and it wants to maximize its profits. Which security should it introduce? For which agent is this introduction more valuable? Is the outcome Pareto optimal? Why or why not?

Problem 2

Consider an economy in which there are two dates $t = 0$ and $t = 1$, and two possible states, $i = 1$ and $i = 2$, at $t = 1$, which occur with equal probability, so that $\pi_1 = \pi_2 = 1/2$. In this economy, two "complex" assets are traded. The first sells for price $P_1^0 = 2$ at $t = 0$ and makes payoffs of $Z_1^1 = 2$ in state $i = 1$ at $t = 1$ and $Z_1^2 = 4$ in state $i = 2$ at $t = 1$. The second sells for $P_2^0 = 1.25$ at $t = 0$ and makes payoffs $Z_2^1 = 1$ in state $i = 1$ at $t = 1$ and $Z_2^2 = 3$ in state $i = 2$ at $t = 1$.

a. Use this information to compute the price q^1 at $t = 0$ of a contingent claim that pays off one in state $i = 1$ at $t = 1$ and zero in state $i = 2$ at $t = 1$ and the price q^2 at $t = 0$ of a contingent claim that pays off zero in state $i = 1$ at $t = 1$ and one in state $i = 2$ at $t = 1$.

b. Use your answers from part (a), above, to compute the price P_b^0 at $t = 0$ of a bond that pays off $Z_b^1 = 1$ in state $i = 1$ at $t = 1$ and $Z_b^2 = 1$ in state $i = 2$ at $t = 1$.

c. Use your answers from part (a), above, to compute the price P_s^0 at $t = 0$ of a share of stock that makes no dividend payments, but can be sold for $P_s^1 = 2$ in state $i = 1$ at $t = 1$ and $P_s^2 = 3$ in state $i = 2$ at $t = 1$.

Problem 3

You observe the following assets with the corresponding state-dependent payoffs:

Securities	A	B	C
States	3	7	8
	1	2	9
	7	16	25

Is this market complete?

Is the following market structure complete?

Assets		A	B	C
States	1	1	5	12
	2	3	6	14
	3	12	7	16
	4	15	8	20

If the market is not complete, introduce a derivative security to complete it.

Construct an Arrow–Debreu security by introducing calls or puts on these assets.

Can you reproduce the same structure by using only puts?